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Cabin Resource Management Ltd.

Geospatial Division

Geospatial Analyst

Interview Round 3 - Machine Learning & Data Case Study

Introduction

Treat this interview like a homework assignment. The problem can be approached in many different ways, and there is no single correct solution.

The objective of this assessment is to evaluate your ability to understand an ambiguous geospatial problem, discover and work with public datasets, create a defensible machine-learning dataset, build and evaluate a model, and communicate your technical decisions clearly.

Your response should include a short presentation delivered virtually covering the problem, the data you selected, your data-preparation and modelling approach, key results, limitations, and what you would do next.

You may submit one email to the recruiting team containing as many clarification questions as you like.

The Challenge

Wildfire and forestry planning frequently depend on accurate information about vegetation and forest structure. However, available forest inventories may be incomplete, outdated, collected at different times, or available at different spatial resolutions. At the same time, current information is publicly available from satellite imagery, terrain datasets, forest inventories, disturbance datasets, and other sources.

For this assignment, assume Cabin would like to create a current estimate of forest crown closure, or an equivalent vegetation-density indicator, across a study area in British Columbia. The goal is not to reproduce an existing Cabin workflow, but to show how you would approach the problem from public data through modelling.

No prepared modelling dataset will be provided.

The Question

Design and demonstrate a reproducible workflow that:

- Identifies and retrieves appropriate public geospatial datasets.
- Cleans and standardizes those datasets.
- Creates a machine-learning-ready dataset.
- Trains and evaluates a baseline model and a stronger candidate model.

You may select a reasonably sized study area in British Columbia. It should be large enough to demonstrate a meaningful spatial problem but small enough to run on a personal computer.

Part 1 - Understanding the Problem

Before writing code, briefly explain how you interpret the problem. Discuss:

- What the model should predict and whether you would treat it as regression or classification.
- What could be used as the ground-truth or target data.
- What you would clarify with a forestry professional before beginning.
- Your key assumptions and what could make the result unsuitable for operational use.

We are interested in your reasoning as much as your implementation.

Part 2 - Public Data Discovery

Identify and retrieve the public datasets required for your solution. Use at least two public data sources from different categories, such as forest inventory/vegetation information, Sentinel-2 or Landsat imagery, terrain/elevation, climate, fire history/disturbance, or land-cover information.

Possible sources include BC government data services, federal data services, Copernicus, Landsat, or equivalent public repositories. Document where the data came from, why you selected it, its spatial/temporal characteristics, and how another analyst could retrieve it again. Where practical, data retrieval should be scripted rather than entirely manual.

Part 3 - Data Cleaning and Dataset Creation

Create a model-ready dataset from the source data. Address the issues that are relevant to your solution, including:

- CRS, raster alignment/resolution, and spatial extent.
- NoData/missing values, clouds or unusable imagery, and obvious data-quality issues.
- Temporal differences between source datasets.
- Label quality, sampling, class imbalance, and potential spatial leakage.

Briefly describe the final dataset: number of samples, input features, target variable, missing-data treatment, and how you created the training/validation/test split. Explain any important decisions that could materially affect model performance.

Part 4 - Machine Learning

Build and compare two models:

1. Baseline model

Use a simple model that provides a meaningful reference point, such as linear/logistic regression, a decision tree, or a basic Random Forest.

2. Candidate model

Build a second model that you believe is better suited to the problem. This may be Random Forest, Gradient Boosting, XGBoost, a neural network/CNN, or another justified approach.

Explain why you chose each model, how you split and validated the data, which metrics you used, and what you learned from comparing the results. Because this is spatial data, be prepared to explain whether a conventional random split is appropriate and how you considered spatial leakage.

Deep learning is welcome but not required. A simpler, well-justified and correctly validated model is preferable to unnecessary complexity.

Code Submission Expectations

Submit working, readable Python or R code that another analyst can follow. A production-ready package is not expected, but avoid placing the entire solution in one large undocumented notebook. Include clear setup/run instructions and reasonable organization of the main data and modelling steps.

Optional Bonus - Additional Depth

The following items are optional. They can be used to demonstrate additional depth, but are not required:

- More detailed model interpretation, error analysis, uncertainty, or spatial bias analysis.
- A GIS-compatible prediction layer and/or a final map of the study area.
- A more automated/reproducible pipeline for updating data and retraining the model.
- Additional software-engineering practices such as configuration files, automated tests, CI/CD, logging, or packaging.

Deliverables

Required:

1. Source-code repository or submitted source code.
2. README with setup and run instructions.
3. Data-source documentation and provenance.
4. Environment/dependency definition.
5. Data acquisition/preparation workflow.
6. Final model-ready dataset, or clear instructions to reproduce it.
7. Baseline model.
8. Candidate model and core comparison results.

Optional bonus deliverables:

- Extended model evaluation or interpretation.
- GIS-compatible prediction output.
- At least one final map.

A short presentation is required. You do not need to submit large raw satellite datasets through Git.

Scope

A production-ready wildfire modelling system is not expected. A small study area, a few well-chosen public datasets, a clean dataset, sensible models, and clear technical reasoning will be considered a strong submission.

We would rather see a simple and defensible solution than an unnecessarily complex one.

Final Presentation Discussion

During the final interview, be prepared to discuss:

1. Why you chose your datasets and study area.
2. The most important data-cleaning or dataset-creation challenge you encountered.
3. Why you chose your baseline and candidate models.
4. How you validated the models and handled spatial leakage.

5. Where you would and would not trust the result.
6. What you would do next if this were becoming a Cabin production workflow.

The Ideal Solution

The strongest submission will show that the candidate can turn an unclear real-world question into a technical problem, independently work with public geospatial data, create a defensible ML dataset, select and validate appropriate models, and communicate the limitations of the result. Model accuracy alone will not determine the quality of the submission.